Problem Set 8

Name: Nicolas Iskandar

Collaborators: None

Problem 8-1.
Subproblems: $x(i, j)$: maximum possible profit by selling j barrels to the suffix of buyers i through n, for $i \in \{1, \ldots, n+1\}$, $j \in \{0, \ldots, m\}$

Relate:

$$x(i,j) = \max \begin{cases} p_i + x(i+1,\ j-a_i) & \text{if } j \geq a_i \\ x(i+1,\ j) & \text{always} \end{cases}$$

Topological Order: $i = n+1, n, \ldots, 2, 1$ ($x(i, j)$ only depends on subproblems with strictly larger i, so acyclic)

Base Cases: $x(n+1, j) = -sj$ for all $j \in \{0, \ldots, m\}$

Original Problem: $x(1, m)$; store parent pointers to reconstruct which orders are filled

Time: $(n+1)(m+1) = O(nm)$ subproblems $\times\ O(1) \implies O(nm)$, which is *pseudopolynomial*

Problem 8-2.
Subproblems: $x(i, j, c)$: maximum possible score playing Split Bowling on substring of pins $V[i : j]$ (zero-indexed), where all pins in range must be knocked down if $c = 1$, unconstrained if $c = 0$, for $i \in \{0, \ldots, n\}$, $j \in \{i, \ldots, n\}$, $c \in \{0, 1\}$

Relate:

$$x(i,j,c) = \max \begin{cases} x(i+1,\ j,\ c) & \text{if } c = 0 \\ v_i + x(i+1,\ j,\ c) & \text{if } i < j \\ \max\limits_{k \in \{i+1, \ldots, j-1\}} \left\{ v_i \cdot v_k + x(i+1,\ k-1,\ 1) + x(k+1,\ j,\ c) \right\} & \text{if } i+1 < j \end{cases}$$

Topological Order: increasing order of $j - i$ ($x(i, j, c)$ only depends on subproblems with strictly smaller $j - i$, so acyclic)

Base Cases: $x(i, i, c) = 0$ for $i \in \{0, \ldots, n\}$, $c \in \{0, 1\}$

Original Problem: $x(0, n, 0)$; store parent pointers to reconstruct which pins are knocked down together

Time: $O(n^2)$ subproblems $\times\ O(n) \implies O(n^3)$, which is *polynomial*

Problem 8-3.

Subproblems: $x(k, s_1, s_2, s_3)$: True if it is possible to partition the suffix of items $A[k:]$ into four subsets A_1, A_2, A_3, A_4 such that $\sum_{a_i \in A_j} a_i = s_j$ for all $j \in \{1, 2, 3\}$, and False otherwise, for $k \in \{0, \ldots, n\}$, $s_1, s_2, s_3 \in \{0, \ldots, m\}$

Relate:

$$x(k, s_1, s_2, s_3) = \text{OR} \begin{cases} x(k+1,\ s_1 - a_k,\ s_2,\ s_3) & \text{if } a_k \le s_1 \\ x(k+1,\ s_1,\ s_2 - a_k,\ s_3) & \text{if } a_k \le s_2 \\ x(k+1,\ s_1,\ s_2,\ s_3 - a_k) & \text{if } a_k \le s_3 \\ x(k+1,\ s_1,\ s_2,\ s_3) & \text{always} \end{cases}$$

Topological Order: $k = n, n-1, \ldots, 1, 0$ ($x(k, s_1, s_2, s_3)$ only depends on subproblems with strictly larger k, so acyclic)

Base Cases: $x(n, 0, 0, 0) =$ True; $x(n, s_1, s_2, s_3) =$ False for any $s_1, s_2, s_3 > 0$

Original Problem: Let

$$m(0, s_1, s_2, s_3) = \begin{cases} \max\{s_1,\ s_2,\ s_3,\ m - s_1 - s_2 - s_3\} & \text{if } x(0, s_1, s_2, s_3) = \text{True} \\ \infty & \text{otherwise} \end{cases}$$

Answer is $\min_{s_1, s_2, s_3 \in \{0, \ldots, m\}} m(0, s_1, s_2, s_3)$; store parent pointers to reconstruct each subset

Time: $O(m^3 n)$ subproblems $\times\ O(1)$, plus $O(m^3)$ to compute the original answer $\implies O(m^3 n)$, which is *pseudopolynomial*

Problem 8-4.

Preprocessing: Scan the $O(m^2n)$ characters across all uncorrupted logs in L_u, inserting each word w encountered into a hash table H mapping to its frequency $f(w)$, in expected $O(m^2n)$ time.

Subproblems: $x(j)$: maximum $\sum_{w \in R_{i,j}} f(w)$ over any restoration $R_{i,j}$ of the suffix $\ell_i[j :]$ of corrupted log ℓ_i, for $j \in \{0, \ldots, |\ell_i| \leq m\}$

Relate:

$$x(j) = \max_{k \in \{j+1,\ldots,|\ell_i|\}} \left\{ f(\ell_i[j:k]) + x(k) \right\}$$

$$f(w) = \begin{cases} H(w) & \text{if } w \in H \\ 0 & \text{otherwise} \end{cases}$$

Topological Order: $j = |\ell_i|, |\ell_i| - 1, \ldots, 1, 0$ ($x(j)$ only depends on subproblems with strictly larger j, so acyclic)

Base Cases: $x(|\ell_i|) = 0$

Original Problem: $x(0)$, computed for each of the n corrupted logs; store parent pointers to reconstruct a restoration achieving value $x(0)$

Time: $O(m)$ subproblems $\times$ $O(m^2)$ work each (expected, due to hashing substrings) $\implies$ expected $O(m^3)$ per log, $O(m^3n)$ total across all logs (plus $O(m^2n)$ preprocessing), which is *polynomial*